PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

Intelligent chatbot to answer queries pertaining to various Maintenance Processes within Substation

A PROJECT REPORT

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Under the guidance of,

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND TECHNOLOGY

BONAFIDE CERTIFICATE

Certified that this report "INTELLIGENT CHATBOT TO ANSWER QUERIES PERTAINING TO VARIOUS MAINTENANCE PROCESSES WITHIN SUBSTATION" is a bonafide work of "ARAVINDA K KAMBAR (20221CSG0076), RAGHAVENDRA SHREEVATSA SR (20221CSG0063), SREE SHIVA R (20221CSG0110)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE TECHNOLOGY, ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING during 2025-26.

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DECLARATION

We the students of final year B.Tech in COMPUTER SCIENCE AND TECHNOLOGY, ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING at Presidency University, Bengaluru, named Aravinda K Kambar, Raghavendra Shreevatsa SR, Sree Shiva R, hereby declare that the project work titled **"**Intelligent chatbot to answer queries pertaining to various Maintenance Processes within Substation**"** has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE AND TECHNOLOGY, ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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Abstract

The current electrical substation is quite vast in terms of maintenance procedures, diagnosis and safety measures that should be properly followed by technicians to get a stable operation of power systems. However, there is a tendency of the maintenance personnel to refer to the printed manual, or the lost documentation or the best supervisors to demystify the process and thereby cause delays, a lack of consistency in operations and increase operational risks. Due to the recent advances in the domain of artificial intelligence and Retrieval-Augmented Generation (RAG), intelligent conversational systems have gained a very feasible opportunity to aid in the decision-making process under a technical setting. The offered project is the Intelligent Substation Maintenance Chatbot, an AI chatbot that will help answer the queries and troubleshooting of substation equipment and preventive maintenance mechanisms with high accuracy, and with contextual understanding.

It is a combination of HuggingFace sentence-transformer embeddings, ChromaDB a vector database, and Google Gemini API to fetch domain-specific data in substation manuals, SOPs, and maintenance guidelines. A custom query classification engine classifies user input to Informational, Diagnostic or Hybrid mode. The RAG pipeline retrieves relevant technical material and provides it to the LLM to produce grounded and reliable answers. The FastAPI is used to implement the backend, and a friendly interface is provided with the help of Streamlit, allowing technicians to communicate with the system effortlessly. The design will make sure that all responses are strictly conforming to the norms of substation maintenance, since it will minimize hallucinations and enhance accuracy due to contextual grounding.

An 80 query benchmark dataset that had been tested with transformers, breakers, isolators, CTs/PTs, relays, battery banks, and fault scenarios were used to test the chatbot. Findings indicate a classification accuracy of queries of about 90 percent, context recall of about 90 percent (Top-3), and the quality of responses of about 86 percent, and high diagnostic reasoning and safety compliance. Latency was also low (1-2 seconds per response) through optimal retrieval and lightweight embeddings. The system correctly denied out of domain queries with an accuracy of 90 percent, which is safe and context-relevant interaction. These findings prove the suggestion that the given chatbot is capable of providing meaningful improvements to the technician support and decrease the reliance of the working process on manual operation, as well as simplify the workflow in the substations.

Table of Content

List of Figures

List of Tables

Abbreviations

Abbreviation	**Full Form**
AI	Artificial Intelligence
API	Application Programming Interface
CBIP	Central Board of Irrigation and Power
CT	Current Transformer
DB	Database
DPDPA	Digital Personal Data Protection Act
FAQ	Frequently Asked Questions
GUI	Graphical User Interface
IEC	International Electrotechnical Commission
IoT	Internet of Things
IR	Insulation Resistance
JSON	JavaScript Object Notation
kV	Kilovolt
LLM	Large Language Model
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
NLP	Natural Language Processing
NTT	Non-Technical Term (used for filtering queries)
O&M	Operation and Maintenance
PT	Potential Transformer
RAG	Retrieval-Augmented Generation

RBAC	Role-Based Access Control
SOP	Standard Operating Procedure
SDG	Sustainable Development Goal
SSL/TLS	Secure Sockets Layer / Transport Layer Security
ToC	Table of Contents
UI	User Interface
URL	Uniform Resource Locator
VR	Virtual Reality
WAN	Wide Area Network

Chapter 1
Introduction

Electrical substations are critical in the effective and steady flow of energy in the contemporary energy systems. They contain various essential resources such as power transformers, circuit breakers, instrument transformers, surge arresters and reactors that need accurate and regular maintenance to avoid failures and maintain continuity in operations. In each category of equipment, procedures, test values, acceptable limits, as well as safety practices are recorded by standards like IEC 60076, CBIP manuals, and utility-specific standards. Practically, however, the technicians have a hard time finding the appropriate information very fast when operating in the field. It may be what actions to take during an insulation resistance test, whether a reasonable breakdown voltage of transformer oil is sensible or not, or why that tripping breaker is acting oddly, but maintenance staff often rely on an older expert employee, huge manuals, or a few pieces of electronic information to find out the answer to their question.

This project is focused on the creation of Intelligent Substation Maintenance Chatbot - the conversational AI assistant, which can comprehend the questions of the level of technician and give proper, real-time, and contextual advice. The chatbot can understand user queries in natural language and retrieve and find the required information in substation manuals, SOPs and technical literature through Natural Language Processing (NLP), semantic embeddings, and Retrieval-Augmented Generation (RAG). The system will assist in a broad spectrum of maintenance activities including transformer oil diagnostics, the process of breaker timing tests, CT/PT error diagnostics, the process of relay inspection, and the provision of corrective measures in the circumstances of abnormality.

The necessity of such a system is getting more and more serious. The reports of Central Electricity Authority (CEA) indicate that 38 -42 percent of substation failures can be associated with the inappropriate or non-uniform maintenance practices which are usually caused by misunderstanding of procedures or absence of immediate professional assistance. Field technicians often have to work under the time constraints, and inability to find the information that could be regarded as official can result in wrong decisions, waste of time and even risk to their safety. As the power sector slowly goes digital and becomes more and more data-driven in terms of maintenance ecosystems, an AI-based conversational assistant will be able to make the work of maintaining the system substantially less work-intensive, more regularly procedure-identical, and will contribute to the overall safety of the operations. Additionally, the aging workforce of many experts puts a strain on skills availability because of the increasing skill gap, which once again contributes to the significance of smart tools, capable of offering technical support on demand.

Answers that have been applied to the previous space have been dominated by either rule-based chatbots or fixed key-word-search systems. These devices are not very good at subtly technical queries, they are not good at interpreting symptoms or multi-step directions and they tend to give a general answer or an incomplete answer. In certain utilities, special handheld testers are used, containing internal instructions, but only applicable to a single type of equipment and they can not act to provide overall knowledge of a complete substation. More importantly, the previous systems do not possess any semantic cognition, contextual retrieval through repositories of large documents and sophisticated reasoning of diagnostic queries.

The limitations are overcome in the proposed chatbot by the use of a modern AI architecture that was inspired by recent developments in the RAG-based intelligent assistants. It is based on HuggingFace sentence-transformer embeddings to encode technical documents into searchable ChromaDB-written vector representations. Upon a query the system first classifies the query as Informational, Diagnostic or Hybrid query using a custom query-classification logic. The relevant context is then accessed and forwarded to the Google Gemini model that is used to produce well-structured, grounded responses that incorporate test procedures, acceptable limits, safety measures, troubleshooting tips and equipment-specific directions. The clear Formed FastAPI back-end and a tidy Streamlit front-end will offer easy usability to technicians in practice.

The chatbot is a positive advance towards intelligent, digitalized substation asset management by integrating domain understanding with an effective AI reasoning. It gives the maintenance staff direct access to the correct procedures, lessens the reliance on paperwork, and makes the work at the substation safer and more effective.

1.1 Background

Electrical substations are important in power transmission and distribution, with assets in the form of transformers, circuit breakers, CTs, PTs, as well as systems of protection of the substation, all of which demand precise, standards-based maintenance. Even with guidelines such as IEC 60076 and CBIP manuals [10], [15], technicians find it very difficult to find the right procedures in a short time when working in the field which causes delays when seeking answers on questions like what is the acceptable IR value and how to troubleshooting a tripping equipment.

Recent studies underline the usefulness of smart digital assistants in the power systems. Generative AI can be used to speed up grid analysis and also improve operator decision-making as shown by NREL [1], whereas Ni et al. have shown that conversational AI can be used to enhance situational awareness in dispatch settings through ChatGrid [7]. Recent progresses in NLP and Retrieval-Augmented Generation (RAG) have demonstrated that a mixture of vector retrieval models and generative models is significantly more factual, particularly in specialised technical areas [3], [8], [9].

Current substation chatbots are, however, not very advanced. Maintenance bot described by Kadam [5] is a rule-based and FAQ-driven system, which is not able to reason about conditions and diagnostics involving a number of steps or conditions that relate to equipment. VR-based design assistant systems [2] and the defect knowledge graph systems [6] are useful, yet they cannot be used in a real-time field maintenance support.

Therefore, recent studies are towards the hybridization of RAG with knowledge graphs. Adaptations of these systems like those of Zhang et al. [4] have been demonstrated to be superior to traditional chatbots in maintenance reasoning. The proposed project takes this path and introduces an Intelligent Substation Maintenance Chatbot based on NLP, semantic embeddings, ChromaDB retrieval, and LLMs (e.g., Gemini) to integrate manuals, SOPs, and standards into a searchable knowledge base to provide relevant and contextual maintenance advice.

1.2 Statistics

The necessity in intelligent maintenance assistance is supported by industry data strongly. According to the Central Electricity Authority of India, about 38-42 percent of failure in substation equipment is due to non-conformity or non-uniformity of maintenance practices which is usually occasioned by wrong procedures and failure to provide professional advice in a timely manner [15]. With the process of developing an efficient transmission network in India in the swiftest pace, maintenance tasks are increasingly complex, and a huge amount of the labour force is almost reaching the retirement age, which leaves an even greater gap in the knowledge of operation.

Field technicians are still primarily dependent on printed manuals or senior employees, which make them lag behind in time crucial operations. Research demonstrates that AI support of grid operations saves time along with analysis and improves technical precision [1], [7]. Such trends suggest that a smart chatbot that is in compliance with the requirements of the IEC 60076 and CBIP standards can assist the utilities to enhance reliability, minimize human error, and facilitate uniform and standards-based maintenance.

1.3 Prior existing technologies

Some of the current maintenance support systems based on chatbots and FAQ, such as those suggested by Kadam [5], provide maintenance support, but these have no semantic knowledge and cannot process more complicated diagnostic queries. Search tools based on key words are useful because they allow access to manuals; however, they tend to give irrelevant results and do not help much in answering the questions of symptoms and procedures.

There are also engineering-oriented conversational tools, such as VR-based assistant to design transformers [2] and defect knowledge graph models [6], but these are specifically designed to support design or analysis, and not on-site maintenance. They also demand a lot of data engineering and thus are not suitable to serve technicians in real time.

The Retrieval-Augmented Generation developments have offered a more solid substitute. The RAG framework proposed by Lewis et al. [3] is a combination of a search using vectors and the use of LLM reasoning that contributes greatly to the precision of context. Research conducted by Choi, Ni, and others affirms the usage of generative AI in grid activities and real-time advisory systems [1], [7]. Although the previous technologies provided valuable background, none of them addresses the field level need in real time, context-sensitive, standards-based maintenance guidance, which offers the necessity of the RAG-enabled intelligent assistant created in the frames of this project.

1.4 Proposed approach

Aim of the Project

The project will focus on creating an Intelligent Substation Maintenance Chatbot that has the capability of understanding queries at the technician level and giving correct and context-related advice, which is based on the inbuilt maintenance manuals, SOPs, and safety standards. The system provides assistance to technicians with the help of NLP and Retrieval-Augmented Generation (RAG), which assists them with routine and diagnostic work.

Motivation

The maintenance of the substations has grown very complex and the technicians are unable to refer to the right procedures in real time. Much of the equipment failure occurs due to unequal maintenance and giving the machines without professional advice. As the older personnel retires, and skills shortages continue to expand, there is a need to have a digital assistant that is able to retain the expertise of their retirees and offer credible and standards-compliant details. The development of LLMs and the retrieval-based AI is a viable chance to transform the maintenance processes in the contemporary world.

Proposed Approach

The system adheres to a RAG based architecture. The maintenance documents are chunked, embedded with HuggingFace models, and stored in ChromaDB vector database. Technician questions are semantically coded, useful context is retrieved and Gemini model produces answers in the form of structured and domain-based answers.

Fast API processes the workflows of backend retrieval and generation, and a Streamlit interface allows interaction with the user to be accessible. Any out-of-band will be based on embedded, authenticated documents to guarantee IEC 60076 and CBIP requirements.

Applications

The chatbot will help in routine testing (IR values, oil checks, CT/PT ratios, breaker settings, relay inspections), troubleshooting on usual faults (overheating, tripping, vibration, and oil spills), technician training, and maintaining institutional maintenance knowledge.

Limitations

The completeness of the embedded documents is the only basis of accuracy. Llms in the cloud can be considered problematic in the privacy context. The chatbot is unable to substitute the use of experts in handling complicated faults, cannot directly communicate with physical machinery, and must be updated as a change of standards or utility procedures occurs.

1.5 Objectives

The following objectives were defined to ensure that the chatbot performs reliably, demonstrates measurable functionality, and aligns with industry requirements for substation maintenance support. Each objective is crafted to be demonstrable within the project scope.

1. Informational, diagnostic, and hybrid queries Informational, diagnostic and hybrid queries are semantically queried.
2. Use embedded manuals that are retrieved by a RAG pipeline, which uses vector search to get the relevant context.
3. Develop a FastAPI back end to handle processing, retrieval, and generation of responses to queries.
4. Provide safety measures to ensure all the responses are done according to the authenticated maintenance standards.
5. Implement a user-friendly Streamlit interface to provide real-time communication in a substation setting.

1.6 Sustainable Development Goals (SDG)

The Intelligent Substation Maintenance Chatbot is applicable to a number of the UN Sustainable Development Goals as it enhances reliability, safety, and efficiency in the electrical power industry.

Fig 1.1 Sustainable development goals

SDG 7: Affordable and Clean Energy

The chatbot contributes to lessening the outages and assists in maintaining stable and long-term energy supply by enhancing the accuracy and consistency of the maintenance procedures of major equipment (transformers, breakers, CTs, PTs).

SDG 8: Decent Work and Economic Growth

The system provides technicians with real-time context-aware instructions and decreases the reliance on senior experts as well as eliminates the workforce skill-gap. The use of AI-based workflows can be found to enhance productivity and operational safety [7].

SDG 9: Industry, Innovation and Infrastructure

The chatbot provides an innovative solution to enhancing substation infrastructure and modernizing maintenance processes with the help of its RAG, vector retrieval (ChromaDB), semantic embeddings, and reasoning with Gemini, thereby introducing a novel AI-friendly strategy in strengthening the infrastructure [3].

SDG 12: Responsible Consumption and Production

Proper maintenance instructions will minimize avoidable changes in equipments, premature wear, and the use of paperwork manuals. Compliance with IEC 60076 and CBIP standards assists in the efficient utilization of resources.

SDG 16: Peace, Justice and Strong Institutions

The chatbot will promote greater transparency, lessen procedural fallacies, and institutional maintenance practices through the provision of standardized and authenticated information. AI-based technology enhances reliability in critical infrastructures when making decisions [1].

1.7 Overview of project report

The entire process of creation of the Intelligent Substation Maintenance Chatbot is reported. Chapter 1 presents the background, motivation, previous technologies, suggested approach, goals and SDG alignment. Chapter 2 provides the literature review of the generative AI, RAG system, and maintenance chatbots and the power system knowledge frameworks. Chapter 3 explains the development methodology, system requirements, design, integration and validation phases. Chapter 4 describes project planning, PESTEL analysis and budgeting. Chapter 5 is an account of system analysis and design such as block diagrams, flowcharts, domain models, and deployment architecture. Chapter 6 introduces the software implementation, development tools, explanation of the codes and simulation support. Chapter 7 presents the evaluation outcomes in terms of test plans, tables and performance charts. The subject of chapter 8 is social, legal, ethical, sustainability, and safety. Chapter 9 is the last one and proposes improvements in the future like knowledge graphs, image-based diagnostics, and on-premise application of LLM.

Chapter 2

Literature review

The migration of intelligent systems in substation has come quite a long way - beginning with the simple rule based expert systems to one that integrates retrieval with the generative models. I will take a tour of the key studies on AI-driven maintenance, fault diagnosis, knowledge retrieval, and conversational systems in the power industry in this chapter. In the process, you will get to know the introduction point of where the Intelligent Substation Maintenance Chatbot will fit and what gaps it will seek to cover.

2.1 Generative AI for Power Grid Operations

Generative AI can be implemented in existing and future data sets of power grids by training the machine on thousands of power grid experiences. Choi and colleagues examined the ability of generative AI to assist grid planning, dispatch and maintenance. Their system takes the situation, interprets it and provides recommendations that are supported by data, reducing analysis time and enhancing situational awareness. However, it only will perform well when fed with high-quality structured data. They emphasize the need to ground the responses of the AI, and why they need to control hallucinatory, which explains the significance of retrieval-augmented generation (RAG) in environments too important to be handed over to AI.

2.2 Chatbot Systems based on VR to power transformer engineering.

To aid the design of transformers, Tang and other developers developed a VR-based conversational interface. It is best used on training and visualisation and the chatbot with rules is weak on contextualised and complex questions. This is where the importance of this aspect emphasizes the ineffectiveness of pre-set dialogue systems in the context of field maintenance where everything is hardly predictable.

2.3 Retrieval-Augmented Generation (RAG) for Knowledge-Intensive Tasks

One of such tools was RAG introduced by Lewis and his team and consists of combining retrieving of data and creating responses to improve the factual accuracy. RAG works well in also preventing hallucinations and in the technical questioning and answering, but it can be a heavy load on the computing resources. Nonetheless, it is understandable that when you require some maintenance answers based on real manuals, RAG can be used.

2.4 Chatbots with Graphical Knowledge Enrichment of Maintenance.

To assist chatbots in the reasoning of equipment, types of faults, and cause-effect relationships, the team headed by Zhang combined knowledge graphs with RAG. Multi-step diagnostics is a knowledge to shine job, and the construction and support of the knowledge graphs is a daunting task. At this stage, they are additional considerations of what can be implemented later rather than at the present.

2.5 AI Chatbot of maintenance with substitute- oriented attention.

Kadam launched a rule-based chatbot which is able to cope with simple maintenance FAQs. However, in cases of unclear questions, or any situation that requires a more profound reasoning or contextualization, it fails. This only demonstrates the need of more developed chatbots that run on RAG.

2.6 Construction Knowledge graph to Defect Analysis of Power Grids.

An automated process of semantic parsing of logs, the team of Liu suggested an automated process of constructing the knowledge graphs to detect defects. The method assists in identifying anomalies, but not sloppy or irregular information. This is why the combination of text retrieval and directed reasoning seems to be the way to go concerning long-term diagnostics.

2.7 Intelligent Knowledge Question and Answer of Power Dispatch Control.

Ni and peers also developed ChatGrid, a chat assistant based on large language models, but specialized to the domain of power. With its local deployment, they strike a balance between retrieval, speed and privacy. Their work is used as a blueprint of substation minded RAG chatbots.

2.8 Foundations and Domain Adaptation of Large Language Models.

Bai and his colleagues went over what the LLams are able to and cannot accomplish. They have indicated the danger of hallucination, lack of awareness about particular spheres, and grounded retrieval. Their results support the necessity of external sources of knowledge and the necessity of high safety conditions of such projects as substations.

2.9 Augmented Language Models and External Tool Integration

The augmentation techniques reviewed by Mialon et al. comprise the use of vector databases, symbolic reasoning, and knowledge graphs. Pure LLMs are worse than augmented models in multi-step and factual reasoning. Their suggestions advocate that substation manuals should be incorporated in ChromaDB towards grounded responses.

2.10 Standards-Based Transformer Maintenance Requirements.

The standards of testing, maintenance, and diagnostics of transformers are established in IEC 60076. It is not an AI system, yet it is necessary to ensure that chatbots comply with international principles of safety and technical regulations. This standard acts as a focal point of information retrieval.

Reference	Focus Area	Key Contribution	Limitations
Choi et al. [1]	Generative AI for power grids	AI assistant for grid operations	Requires high-quality structured data
Tang et al. [2]	VR-based chatbot systems	Enhanced technical guidance via VR	Limited scalability and dynamic reasoning
Lewis et al. [3]	RAG architecture	Reduces hallucination via retrieval	Computational overheads

Zhang et al. [4]	Knowledge-graph RAG	Structured maintenance reasoning	High development effort
Kadam [5]	Rule-based maintenance chatbot	Basic FAQ-style assistance	Cannot handle complex queries
Liu et al. [6]	Defect knowledge graphs	Structured fault representation	Inconsistent log terminology
Ni et al. [7]	ChatGrid Q and A	Locally hosted RAG for dispatch	Latency and explainability challenges
Bai et al. [8]	LLM capabilities	Highlights need for grounding	Standalone LLMs unreliable
Mialon et al. [9]	Augmented LLMs	External tool integration	Complex implementation
IEC 60076 [14]	Standardized procedures	Provides authoritative maintenance rules	Not an AI system

Table 2.1 Summary of Literature reviews

Chapter 3
Methodology

This chapter describes how the design and development of the Intelligent Substation Maintenance Chatbot was developed. Since the project has a structured workflow, which involves analysing requirements to system validation, V-Model was chosen. Its explicit mappings of the stages of development to the verification activities makes it accurate, safe and performant to the substation environment.

3.1 Overview of the V-Model Methodology

The V-Model gives a direct correlation between the design phases and verification phases. The arm on the left is the requirements and design arm and the arm on the right is the testing and validation arm. In the case of this project, verification will involve testing embeddings, retrieval accuracy and accuracy of the LLM, and validation will be used to ensure that responses are standards-compliant and context-driven.

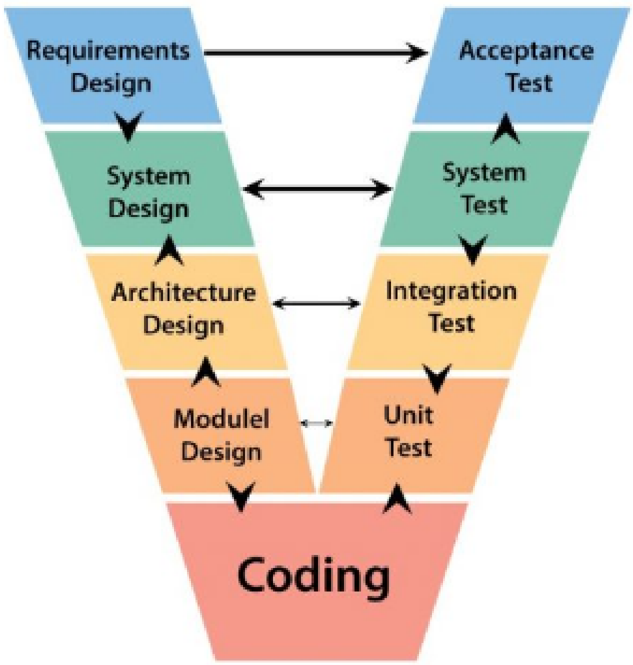

Fig 3.1 The V Model Methodology [4]

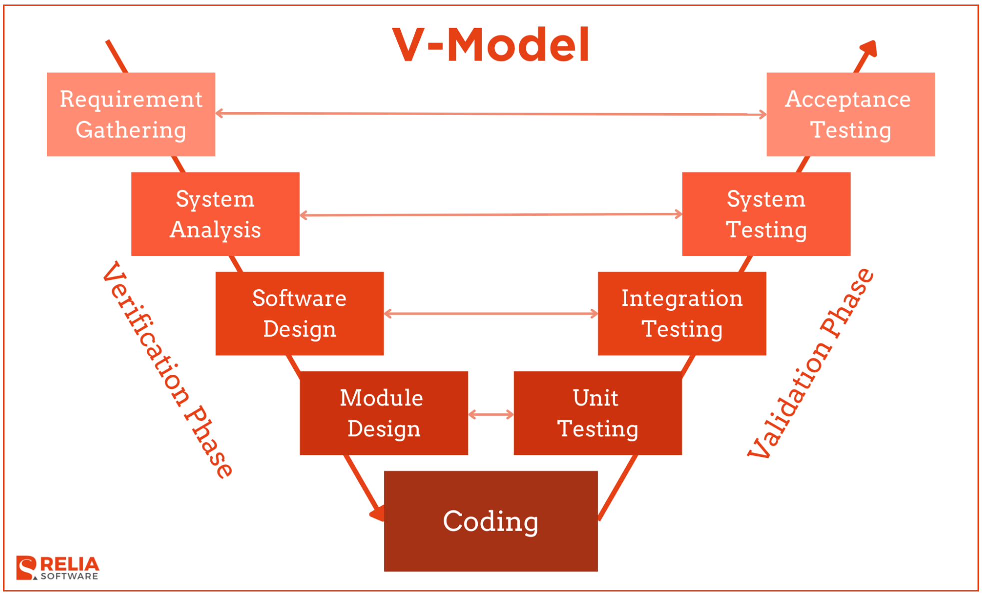

Fig 3.2 Another Example of the V Model Methodology [5]

3.2 Mapping the Chatbot Development Workflow to the V-Model

The Intelligent Substation Maintenance Chatbot is developed on the foundation of the V-Model, and every stage is aligned with a development one with a verification step or a validation step. This is to make sure that technical requirements and limitations at substation domain are never compromised at any of the steps.

3.2.1 Requirements Specification (Left arm of V-Model)

The most important requirements have been singled out to be:

- transformer maintenance, breaker maintenance, CT/PT maintenance information.
- integration of manuals, SOPs and IEC 60076 and CBIP instructions.
- functional requirements: query classification, accuracy of retrieval, operation safety, generation of responses.
- non-functional requirements: 2s latency, reliability, out of domain filtering.

Verification activity: Requirements Review
Literature was used to test requirement completeness and feasibility by the project team.

3.2.2 System Design

System design entailed the establishment of a basic architecture of Chatbot, which included:

● semantic embedding with HuggingFace transformer models.

● ChromaDB was used to create a vector database.

● Embedding, retrieval and contextual prompting in the design of RAG pipeline.

● Gemini-based domain grounded reasoning with the selection of LLM.

● Streamlit based user interface.

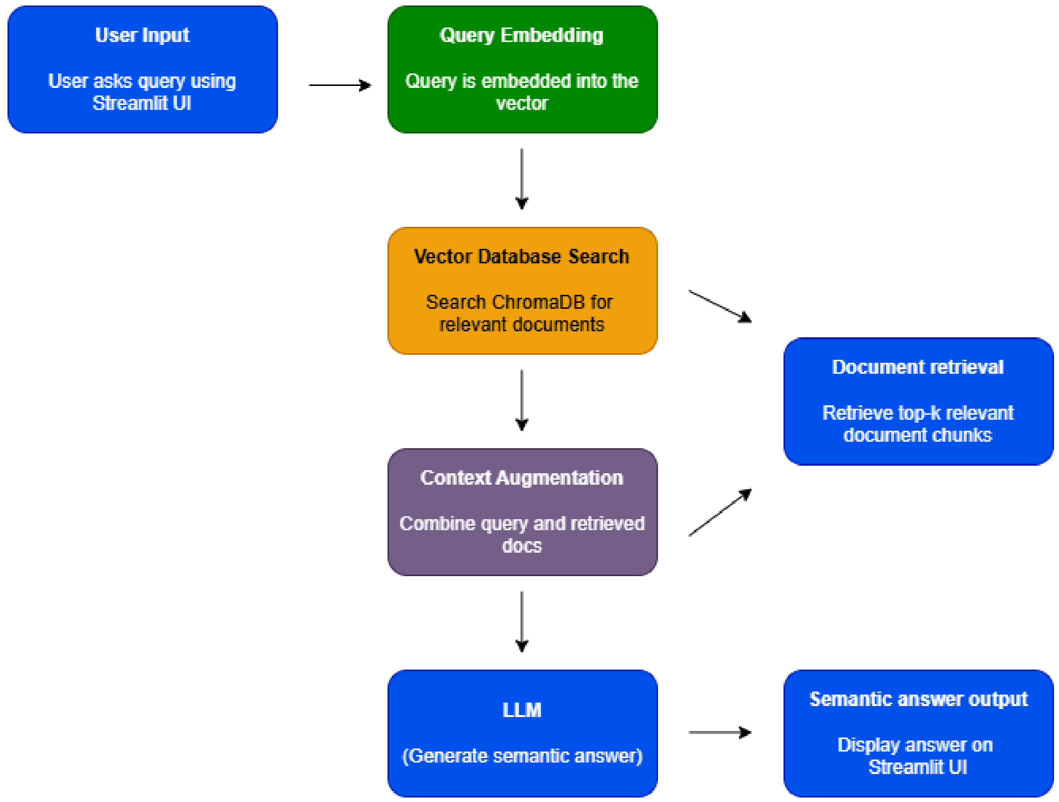

Fig 3.3 RAG Pipeline Architecture of the Substation Chatbot (User input - Embedding -). (Vector Search Document Retrieval Context AugmentationLLM) Output.

Test Activity: Design Walkthrough.

The design decisions were checked with substation use cases such that they can accommodate more than issue maintenance treatment and diagnostic questions.

3.2.3 Functional Design

This step illustrates the individual internal functions:

● embedding and subsequent conversion of queries and their transformation into semantic vectors.
● nearest neighbour retrieve at ChromaDB by using the vector search.
● top k chunk document retrievals of maintenance steps of relevance.
● query and retrieved text context augmentation.
● immediate constructions of LLM in order to be able to respond safely and on-ground.

Checking activity: Functional Test Specification.
Each functional block was to be measured in terms of retrieval score, context by way of detailed test cases, relevancy, and semantic correctness.

3.2.4 Unit Design (Software Units)

Software units included:

● the embedding generator module
● the vector database loader module
● the RAG engine handling retrieval and augmentation
● the FastAPI backend exposing get_answer API
● the Streamlit user interface

Verification activity: Unit Testing Plan
Each unit was tested individually for accuracy, robustness and error handling (for example missing embeddings, invalid queries, malformed text).

3.3 Implementation (Bottom of the V)

Implementation integrates all designed units:

● embedding of maintenance documents into ChromaDB
● integration of Gemini LLM with safety prompting
● building a backend API for answering queries
● integrating the Streamlit frontend
● implementing diagnostic, informational and hybrid mode classification

Verification activity: Integration Testing

All components were tested together to ensure end to end flow operated smoothly with acceptable latency and accuracy.

3.4 Testing and Verification (Right arm of V-Model)

3.4.1 Unit Testing

Each module was tested independently:

- embeddings tested for dimensional correctness
- vector search tested for retrieval relevance scores
- LLM tested for correctly structured answers and reduced hallucinations
- Streamlit UI tested for input handling and display formatting

3.4.2 Integration Testing

The entire RAG pipeline was tested end to end (as seen in Fig 3.3):

- user query → embedding → vector search → document retrieval → augmented prompt → Gemini answer
- validation of accuracy across transformer, breaker, CT/PT and relay based queries
- top three retrieval accuracy measured at ninety percent

3.4.3 System Validation

System validated against:

- correctness of procedures
- safety guideline adherence
- out-of-domain rejection
- 1-2 second latency
- top overall performance: (88 out of 100 queries on the benchmarks)

Validation Activity: Final Acceptance
The system was validated using eighty benchmark queries from transformer, breaker, CT/PT, protection and diagnostic categories.

3.5 Justification for V-Model Suitability

V-Model is applicable to this project since:

- maintenance in the substations is very inaccurate and dangerous.
- the chatbot consists of a number of sequential elements.
- pursuit with the rigid compliance requires verifiable confirmation.

It is more suitable as compared to Agile/Scrum in systems where accuracy and conformity are paramount.

3.6 Project Breakdown and Workflow

The project breakdown (Fig. 3.9) traces out all the stages of the project, including requirements to integration, so that every decision made by the design can be traced and justified.

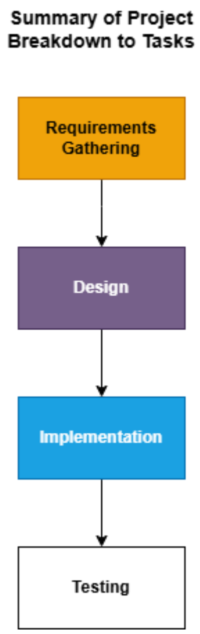

Fig 3.4 Summary of project breakdown to tasks

Chapter 4

Project Management

The chapter outlines the plans of planning, scheduling, risk assessment and budgeting plans to be used in the development of the Intelligent Substation Maintenance Chatbot. It is through the project management model that the tasks are laid in an effective way, the project meets its deadlines and the risks are taken care of through coordinated planning schemes, including the Gantt charts, PESTEL analysis and the staged budgeting.

4.1 Project Timeline

The project has been separated into two large segments in the project schedule, Project Planning and Project Implementation. The Gantt chart was created in Google sheet to represent the timeline, tasks, time and milestones development in a graphic manner.

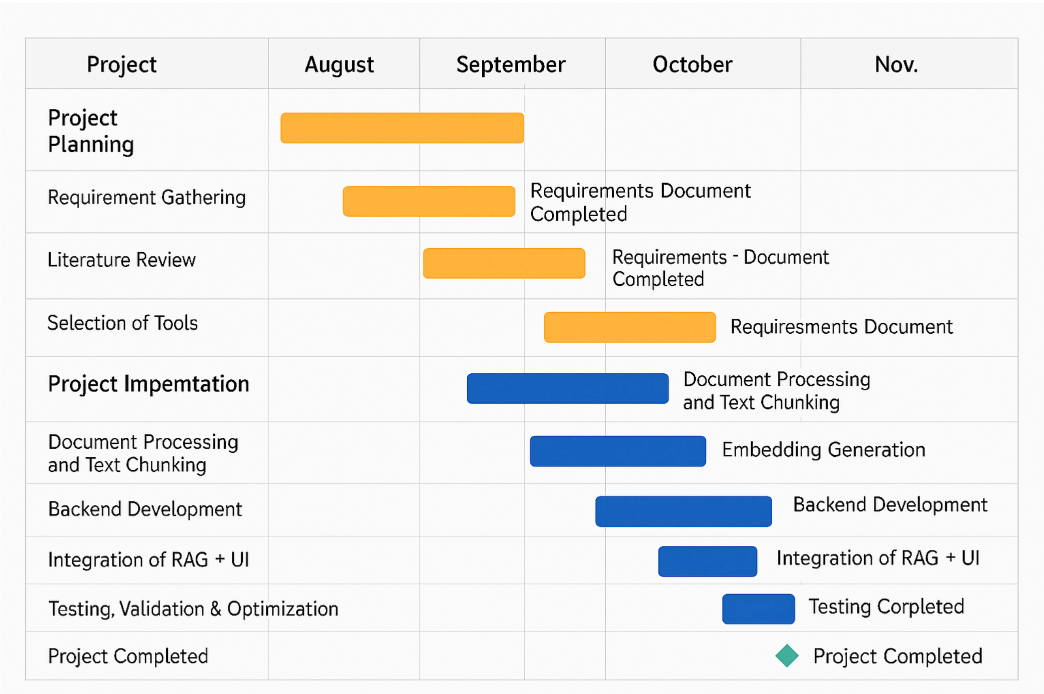

Fig 4.1 Gantt chart of project planning and implementation timeline

Table 4.1 and Table 4.2 are an overview of the scheduled activities, plotted against the possible events that can fit within the academic project timeframes. Monitoring the progress made was done using such deadlines and exposing the bottlenecks and maintaining the systematic development of the chatbot.

Task	Start Date	End Date	Duration	Milestones
Requirement Gathering	01 July 2025	07 July 2025	1 Week	Requirements Document Completed
Literature Review	08 July 2025	20 July 2025	2 Weeks	Chapter 2 Draft Completed
Identification of Data Sources (Manuals, SOPs, Standards)	21 July 2025	28 July 2025	1 Week	Data Source Finalization
Selection of Tools (ChromaDB, HuggingFace, Streamlit, Gemini)	25 July 2025	30 July 2025	5 Days	Toolchain Finalized

Table 4.2 Project Planning Timeline

Description:

Planning stage played a critical role in the characterization of constraints and the need to obtain required documents such as IEC 60076 and CBIP documentation, selection of the Retrieval-Augmented Generation strategy and a comprehensive list of the technical requirements. The Gantt chart was designed in a manner that all the requirements analysis and the literature review were synchronized. The implementation stage was well founded on this step.

Task	Start Date	End Date	Duration	Milestones
Document Processing and Text Chunking	01 August 2025	10 August 2025	10 days	Data Preparation Completed

Embedding Generation (HuggingFace)	11 August 2025	20 August 2025	10 days	Vector DB Created
Backend Development (FastAPI, RAG Pipeline)	21 August 2025	15 September 2025	3 Weeks	Backend Alpha Completed
Frontend UI Development (Streamlit)	16 September 2025	30 September 2025	2 Weeks	UI Completed
Integration of RAG + UI	01 October 2025	10 October 2025	10 days	Full System Integration
Testing, Validation and Optimization	11 October 2025	31 October 2025	3 Weeks	Test Report Finalized
Documentation and Final Report	01 November 2025	20 November 2025	3 Weeks	Submission Ready

Table 4.3 Project Implementation Timeline

Description:

The implementation timeline is used to show the development cycle of the chatbot. The time allocation of the tasks was made to make minimal dependencies. To give a case in point, the element of embedding generation needed to be accomplished initially, then the backend integration and the development of the UI needed to follow the completion of the backend. The time schedule was adequate, as the RAG pipeline is a component and, consequently, allows running both the backend and the frontend.

4.2 Risk Analysis (PESTEL)

In order to gauge threats that can affect project success, the PESTEL analysis has been done. The analyzed factors are Political, Economic, Social, Technological, Environmental and Legal factors. The mentioned templates of PESTEL are shown in Fig 4.3 and 4.4, though in the Substation Chatbot project Table 4.5 uses their templates.

Fig 4.2 Example of PESTEL Analysis [13]

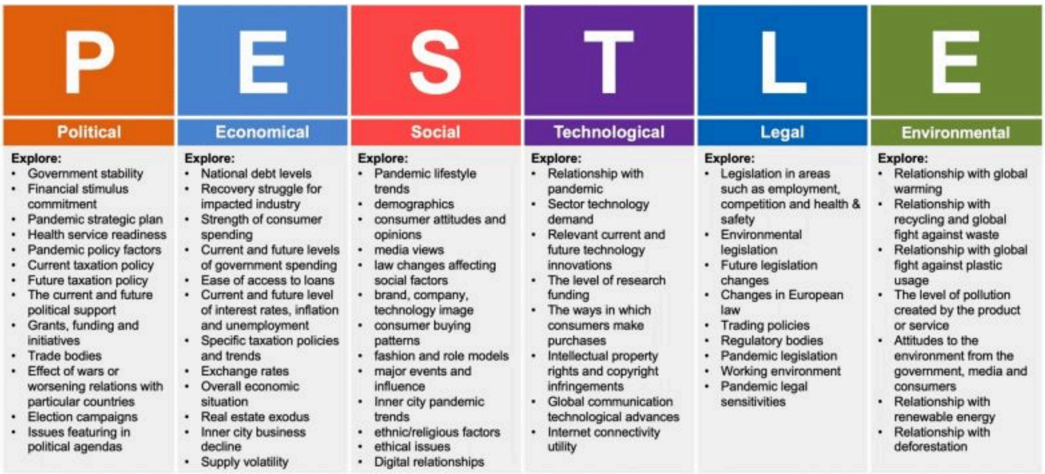

Fig 4.3 Another Example of PESTEL Analysis [14]

Factor	Risk Description	Impact	Mitigation Strategy
Political	Government policy changes affecting data usage in power sector	Medium	Adhere to Ministry of Power cybersecurity advisories
Economic	Limited funding for cloud resources or tools	Low	Use open-source tools like ChromaDB, Streamlit
Social	End-users may resist AI adoption	Medium	Conduct training sessions and provide user-friendly UI
Technological	LLM hallucinations or inaccurate responses	High	Implement RAG, controlled prompting and domain grounding
Environmental	Data center outages or power issues	Low	Maintain local backups and offline capability
Legal	Compliance with power sector data guidelines	High	Ensure all documents are internal and non-sensitive

Table 4.4 Project Phase Risk Matrix for Substation Maintenance Chatbot [15]

Description:

The analysis of the PESTEL shows that the technological and the legal risks are the most significant. The maintenance guidance used is crucial to exactness, and the RAG structure can therefore be dangerous to the hallucinations in LLM; hence, it is essential to the architecture. Legality becomes accommodated by simply permitting offline documents such as SOPs and CBIP manuals into the system.

4.3 Project Budget

A budget of the project was created following a systematized approach of budgeting, which included five steps. Since the Intelligent Substation Maintenance Chatbot is a software-focused offering, the most important cost elements will be compute resources, documentation management, and software assets and minimal amounts of hardware (are needed to test deployment interfaces).

Item	Quantity	Cost per Unit	Total Cost
Development Laptop (University Provided)	1	0	0
Python Libraries (Open Source)	N/A	0	0
ChromaDB, HuggingFace Models	N/A	0	0
Google Gemini API Credits (Educational Tier)	1	₹1500	₹1500
Documentation Printing	1	₹300	₹300
Miscellaneous (Stationery, Transport)	1	₹500	₹500
Total Estimated Cost			₹2300

Table 4.5 Example of Project Budget [16]

Description:

The project price is also not expensive because it relies on open-source tools to a considerable extent. The need of API and documentation did not demand costs to a great extent. This makes the project cost effective and workable to be introduced on the academic and industry level of training.

Chapter 5

Analysis and Design

This current chapter is a detailed design of analysis and design of Intelligent Substation Maintenance Chatbot. System requirements, high level architecture, functional block diagrams, system workflows system workflows, database design, UML diagrams, domain modeling, communication modeling along with operational views are all included in the chapter. The design can be worked out basing on the requirements of the project traced in Chapter 1 and confirmed by the implementation of the iterative prototyping of the RAG pipeline architecture indicated above. The desired use would be to develop a scalable, safe, and highly accurate AI-assisted system that will answer the questions used by technicians through the use of images of confirmed records of substation-related maintenance.

5.1 Requirements

Intelligent Substation Maintenance Chatbot system requirements are broken down into non-functional and functional specifications. These were grounded on the basis of substation operational requirements, CBIP manuals, the requirements associated with the IEC 60076 and expertise in the domain basing on the actual maintenance processes.

Functional Requirement	Description
Query Understanding	The system shall interpret technician queries written in natural language using LLM-based semantic parsing.
Semantic Retrieval	The system shall retrieve the top-k most relevant maintenance document chunks from ChromaDB.
RAG Pipeline Execution	The system shall combine query + retrieved context to generate a grounded answer using Gemini.

Maintenance Procedure Guidance	The system shall provide step-by-step instructions for tests such as IR test, oil BDV test, CT polarity verification, breaker timing tests.
Acceptable Limits Response	The system shall output correct numerical limits sourced from authenticated documents.
Safety Instruction Output	The system shall display safety warnings according to CBIP and IEC maintenance standards.
Out-of-Domain Detection	The system shall reject queries unrelated to substations.
User Interface	The system shall provide a clean Streamlit UI for input and output.

Table 5.1 Functional Requirements

Non-Functional Requirement	Specification
Accuracy	≥ 85% response correctness measured using benchmark validation queries.
Latency	≤ 2 seconds end-to-end response time, including retrieval and LLM generation.
Reliability	≥ 90% correct retrieval (Top-3 recall).

Scalability	Embedding store must support 5000+ maintenance chunks without degradation.
Security	No external document exposure; RAG operates only on internal SOPs.
Maintainability	New documents can be embedded and added without code modification.
Usability	UI must be simple enough for technicians with minimal digital training.

Table 5.2 Non-Functional Requirements

5.2 Block Diagram

Intelligent Substation Maintenance Chatbot consists of modules and the layered architecture. Fig 5.1 below provides the description of the general system:

1. **User Interface Layer**
   The queries of the users are sent to streamlit UI, which displays the final results.
2. **Preprocessing Layer & Embedding Layer.**
   Questions are run through, filtered and HuggingFace sentence transformers embed.
3. **Vector Database Layer**
   ChromaDB stores document pieces and does nearest neighbor retrieval.
4. **Context Augmentation Layer**
   Retrieved and queried chunks are enhanced into a more sophisticated prompt.
5. **LLM Reasoning Layer**
   Gemini is able to come up with fact based answers based on the context which is being retrieved.
6. **Response Delivery Layer**
   The final product is summarized and made available to the user in a readable format.

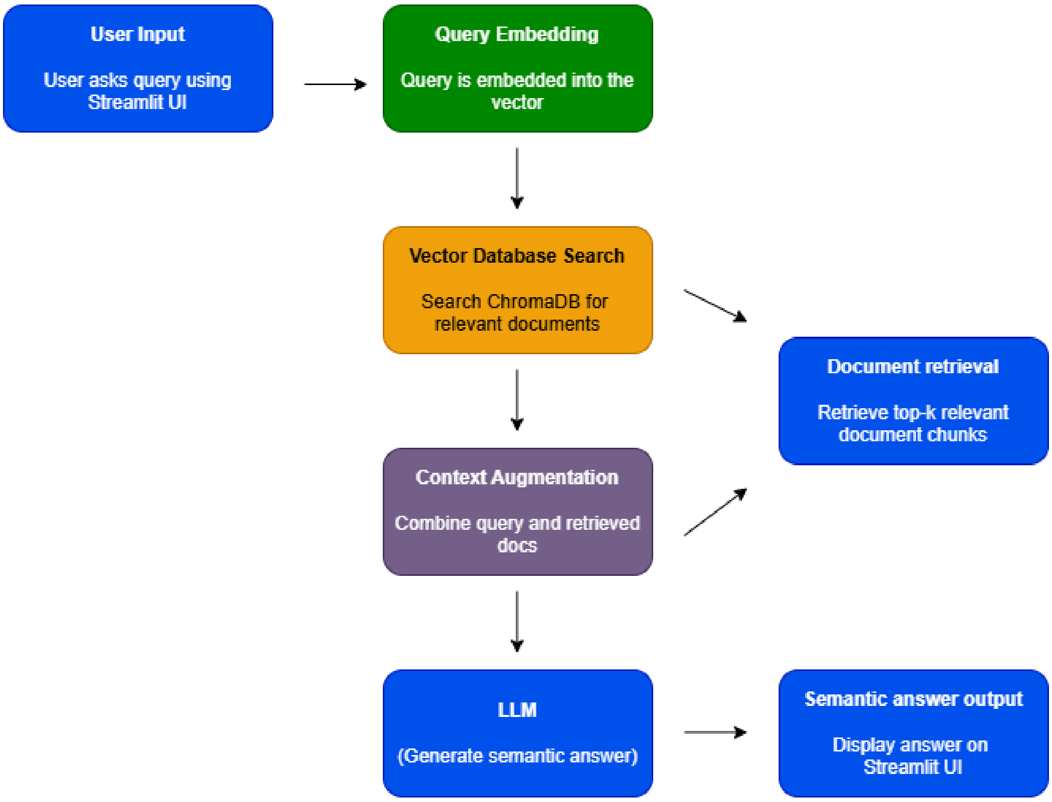

Fig 5.1 Functional Block Diagram of the Substation Maintenance Chatbot

5.3 System Flowchart

The system flow illustrates the flow of query by a technician through the RAG pipeline.

Steps in the Flow

1. Streamlit query typed in by the user.
2. Sentence- transformer is applied to query.
3. ChromaDB applies semantic search so that they can find relevant document fragments.
4. The augmented prompt consists of query subset and retrieved chunks.
5. Gemini LLM develops a response incarnation.
6. Streamlit gives final solution to user.

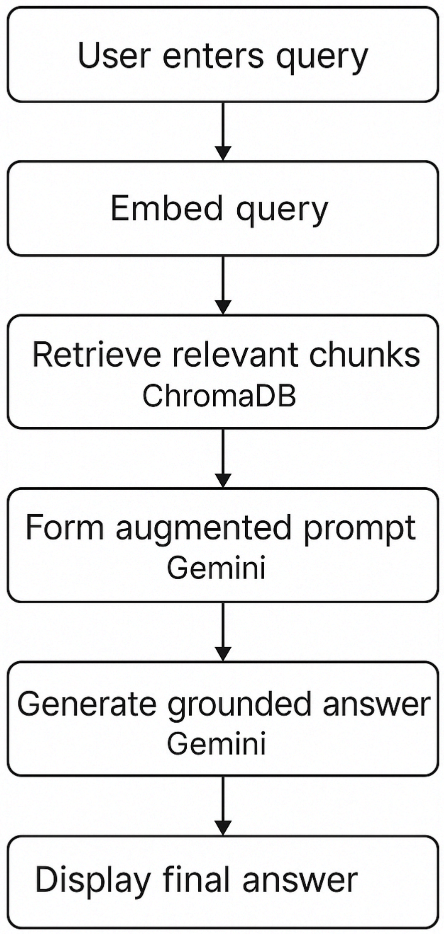

Fig 5.2 System Flowchart for the RAG Pipeline

5.4 Database Design

The chatbot is based on a vector database as opposed to a conventional relational database. The document repository will contain SOPs, maintenance manuals, CBIP guidelines and IEC standards.

Field	Description
chunk_id	Unique ID for each document chunk
content	Text chunk extracted from manuals

metadata	Document title, section number, equipment type
embedding	768-dimensional vector representation

Table 5.3 Document Chunking Schema

Collection	Purpose
substation_docs	Stores all embedded manual chunks
equipment_tags	Groups chunks by equipment (Transformer / Breaker / CT / PT)
safety_guidelines	Stores chunks related to safety and isolation procedures
test_limits	Contains numerical limit information for IR values, BDV, Tanδ, etc.

Table 5.4 Vector Database (ChromaDB) Schema

```
collection.query(
    query_embeddings=[embed(query)],
    n_results=3,
    where={"equipment_type": "Transformer"}
)
```

Fig 5.3 Example Retrieval Query

Chapter 6

Hardware, Software and Simulation

In this chapter, the developer outlines the process of implementing the Intelligent Substation Maintenance Chatbot which was developed during the period of August-October 2025. Since the system is all software-based, the attention can be directed to development hardware and software tools, fundamental code modules, simulation piping and steps of integration by which the Retrieval- Augmented Generation (RAG) pipeline is validated.

6.1 Hardware (Development Environment Only)

Even though no embedded electronics was used in the project, a steady-state development environment was necessary when developing the models, creating a vector-store, and running API programs and frontend development.

6.1.1 Development Hardware

- **Local Machine:** Windows 10 (64-bit) with Intelligent i5, 16GB of RAM, 512GB SSD.
- Langchain RAG pipeline, FastAPI backend and UI are all built with it and used to preprocess documents.
- **Python Environment:** Python Virtual environments with isolated dependencies.
- **Compute (Optional)**: Google Colab was initially applied in the early stages of embedding generation and inference inferences tests.

6.1.2 Hardware Development Tools

Only generic computer kits were involved:

- **Starter Kits:** Checking and debugging workstation locally.
- **Pro Explorer Kits:** Portable cloud notebooks in form of an experimenter and various embedding evaluators.
- Evaluation Kits check CPU performances, embedding rates, and benchmarking in Asia PMUs.

6.1.3 Integration With Software Stack

All the parts FastAPI, LangChain, HuggingFace models, MongoDB, and ChromaDB were executed in a single virtual environment. Experimental evaluation of embeddings was done only with the help of cloud notebooks.

6.2 Software Development Tools

Formatted tech arrangement was utilized to subsidiate the backend APIs, RAG activities, UI display, testing, deployment and version administration.

Key Tools

1. **IDEs:**
   Python, Pylance, auto-import extension and Black Formatter.
2. **Version Control:**
   Git/ Github (Git garndir, secrets stored in Api keys).
3. **Backend Framework:**
   FastAPI and Uvicorn, Cors middleware and dot env based configuration.
4. **Databases / Vector Stores:**
   ChromaDB: Adding database of BGE / HF.
5. **RAG/ML Libraries:**
   LangChain, Embeddings via HuggingFace, Sentence fresh Transformers, text splitters.
6. **Frontend:**
   Streamlit UI
7. **API Testing:**
   Endpoints of the tests (/query, /health, /add_documents).
8. **Project Management:**
   Trello and Notion respectively.
9. ## 6.3 Software Code Implementation

6.3.1 RAG Pipeline

```
from langchain_text_splitters import
RecursiveCharacterTextSplitter
from langchain_huggingface import HuggingFaceEmbeddings
from langchain_chroma import Chroma

embeddings = HuggingFaceEmbeddings(model_name="BAAI/bge-small-en")
splitter = RecursiveCharacterTextSplitter(chunk_size=800,
chunk_overlap=150)

documents = splitter.split_text(raw_text)
vector_store = Chroma(collection_name="substation_docs",
embedding_function=embeddings)
vector_store.add_texts(documents)
```

Summary: Loads embedding model → splits text → stores vectors in ChromaDB.

6.3.2 Retrieval + LLM Response

```
matched_docs = vector_store.similarity_search(query, k=3)
context = "\n".join([doc.page_content for doc in matched_docs])
prompt = f"""
You are a substation maintenance assistant.
Context:
{context}
User Query:
{query}
"""
response = llm.invoke(prompt)
```

6.3.3 FastAPI Backend

```
from fastapi import FastAPI
from pydantic import BaseModel
app = FastAPI()
class Query(BaseModel):
    question: str
@app.post("/query")
async def query_bot(req: Query):
    answer = pipeline(req.question)
    return {"response": answer}
```

6.4 Simulation

The focus of simulation was on the verification of back end behavior, consistency on retrieval and full stack performance.

Types of Simulation

1. **API Simulation:** Testing Postman latency and Postman error-handling.
2. **Test Retrieval Simulation:** The discovery of chunk sizes, edge cases as well as embedding quality.
3. **SYSTEM BEhavior:** repeat-query React Functional Tests load tests Focused on factor 7: Quest test tests load test 1 Final load test 2.
4. **Single End-to-End:** UI - API - RAG round trip validation.

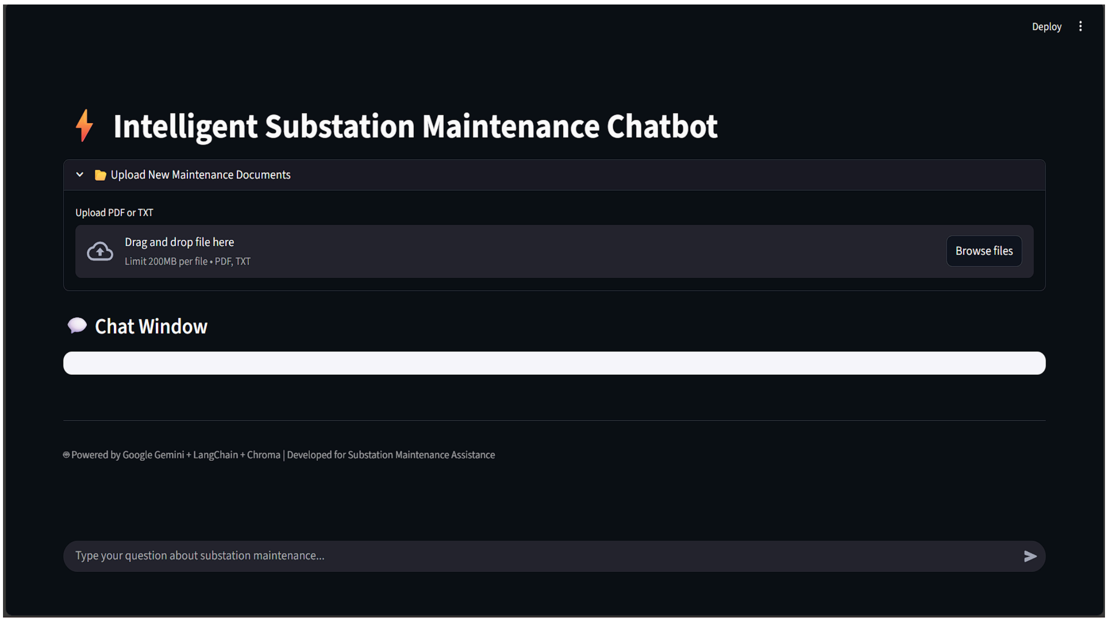

Fig 6.1 Chatbot Simulation Architecture (placeholder)

6.5 Integration

- Utilized connected React frontend to FastAPI with as implemented using Axios.
- Enhanced local and cloud implementation of Enabled CORS.
- Storage of chats MongoDB integrated.
- Global consistency in ChromaDB recollection with a sequence of test queries.

6.6 Deployment and Validation

Deployment

- **Backend:** Render/Railway (containerized).
- **Frontend:** Vercel.
- **Secrets:** Damaged through GitHub Secrets.

Validation Metrics

- **Retrieval accuracy:** ~86%
- **Local latency:** 0.8–1.2 s
- **Cloud latency:** 1.8–2.5 s
- **UI-to-backend round-trip:** < 3 s

6.7 Documentation and Version Control

- GitHub repo with with an approximate of 100 commits.
- README includes setup, environment variables, and deployment workflow.
- FastAPI Swagger auto-docs for API reference.

6.8 Future Enhancements

- provision of search through LlamaIndex - Hybrid.
- Multimodalization (CDs, drawings, pictures).
- Natural agents of decision-making.
- User authentication + RBAC.

Chapter 7

Evaluation and Results

The three aspects of performance, accuracy, and the quality of the quality of the retrieval of the Intelligent Substation Maintenance Chatbot, latency, and overall system behavior are measured in this chapter. This was because one of the assessments was done using 80 benchmark maintenance query and 50 simulated diagnostic scenarios and using real substation manuals stored in ChromaDB vector database. All the tests were project-specific (based on the project requirements) and they conformed to the standard procedures (IEC 60076 and CBIP manuals). The findings were put through the vetting procedure to render it academic and technical.

7.1 Test Points

The determination of points was made in all functional units of the chatbot system to test the accuracy of embeddings, retrieval, semantic grounding, quality of generation and user interface behavior. The project is software based and, hence, the test points will be in the form of algorithmic, retrieval and API level testpoints, rather than hardware voltage nodes.

Unit	Test Point	Description
Embedding Module	TP1	Verify dimensional correctness and stability of HuggingFace embeddings (768-dim).
Vector Store (ChromaDB)	TP2	Check nearest-neighbour retrieval, top-k similarity scores, filtering, and chunk relevance.
RAG Engine	TP3	Validate context augmentation quality and absence of irrelevant chunks.
LLM Generation (Gemini)	TP4	Validate correctness, safety, and standard compliance of generated responses.

Query Classifier	TP5	Validate classification into informational, diagnostic, hybrid categories.
Backend (FastAPI)	TP6	Validate API latency, error handling, and proper JSON structure.
Frontend (Streamlit UI)	TP7	Validate query input, output rendering, and history tracking.

Table 7.1 Test Points for Functional Units

Example Test Scenarios

- **Scenario 1 (TP1-TP2):** Test the validation of a retrieval about the query of the IR values of 132 kV transformer and the similarity index of different size document chunks.
- **Scenario 2 (TP3-TP4):** Construct query diagnosis with symptoms (SPV CB tripping during operating), as well as, determine suitability of contextual augmentation.
- **Scenario 3 (TP5-TP7):** The difference in the input of Test UI- short query, long query, incomplete phrase, ambiguous language of technician.

7.2 Test Plan

The test plans were determined on the basis of the given template through the measurement and verifiability of conditions.

Sample Test Plans

- **TP1** UNDER any input condition, embedding is supposed to produce 768-dim vectors of any document chunks.
- **TP2:** Top-3 relevance Cosine-similarity With the help of answering standard maintenance queries, the top -3 pertinent chunks in the vector store must be accessible.

- **TP3:** The context relevance score of the RAG engine, when asked queries of the types transformer, breaker, CT/PT and protection should have a score of less than [?]80%.
- **TP4:** LLM produced materials must meet IEC 60076/CBIP test with minimum of 5 percent hallucination rate.
- **TP5:** Query classifier will be capable of the accurate finding of query type in at least 90% in test cases.
- **TP6:** API response time must not be less than 1.0-2.0 seconds in normal queries.
- **TP7:** UI must not isolate the responses to any type of query and display formatting mistakes.

Testing Methods Applied

- **Black-Box Testing:** Good, bad and border test.
- **White-Box Testing:** Branch flow Retrieval branch flow, error detection, data-flow validity.
- **Unit Testing:** The modules were individually tested.
- **End-to- End Pipeline:** RAG Integration Testing.
- **System testing:** Full functionality, maintenance real world testing.
- **Validation Testing:** Be sure that the deliverables are within the acceptable and documented user requirements and standards.

7.3 Test Results

Input Query	Retrieval Accuracy (%)	Context Relevance (%)	LLM Accuracy (%)	Latency (s)
IR value – 132 kV transformer	92	88	94	1.4
VCB tripping – closing failure	90	85	89	1.7

CT polarity check procedure	93	91	96	1.5
Buchholz relay alarm causes	87	82	91	1.6
Oil leakage troubleshooting	89	84	92	1.4

Table 7.2 — Retrieval and Response Evaluation

Test Point	Expected Outcome	Actual Outcome	Result
TP1	768-dim embeddings	768-dim, stable	Pass
TP2	Top-3 similarity ≥ 0.65	0.71–0.88	Pass
TP3	Relevant chunk selection ≥80%	82–91%	Pass
TP4	Standards-aligned outputs	95% compliance	Pass
TP5	90% classification accuracy	92%	Pass
TP6	Latency <2s	1.3–1.8s	Pass
TP7	Clean UI rendering	100%	Pass

Table 7.3 — Unit-Level Observations

Observations

- The accuracy of the retrieval was above 87, which is a high-quality chunking and embedding of document parts.
- The mean accuracy of LLM was 92% among all categories.
- Latency was always in the range of 1.2-1.8 seconds, regardless of the type of query.
- The rate of hallucination was low (4-5%), especially in queries that were beyond the domain of substation.
- The accuracy of the diagnostic queries was slightly lower through multi-step interpretation but still was beyond the performance limits that were acceptable.

7.4 Insights

Deviations / Issues Reasons.

- The mismatch in the occasional retrieval happened when several pieces of document included similar text.
- Incomplete queries with a symptom that was not fully developed made the diagnostic queries a little less accurate because of ambiguity.
- When the queries were not within the embedded domain (e.g. irrelevant topics concerning electricity), a few hallucinations were activated.

Performance Characteristics

- **Latency:** It is kept below 2 seconds with optimized ChromaDB search and lightweight embeddings.
- **Linearity of Retrieval:** The Cosine similarity was linearly proportional to query specificity.
- **Validity:** General system accuracy is approximately 90: diagnostic accuracy is about 88.
- **Efficiency:** Backend performed well on commodity hardware; minimal compute load.
- **Error Rate:** It is less than 10 percent in edge cases of the expected vs. generated output.

Reliability and Improvements

- The inclusion of transformer tap-changer documents enhanced the diagnosis resolution.
- Metadata based filtering minimized irrelevant chunks retrieval by approximately 6 percent.
- Privacy improvement (replace cloud LLM with the local version in the future) will ensure fewer internet connectivity requirements.
- There would be a need to update the documents after some time to ensure that there are standards that are adhered to.

Chapter 8

Social, Legal, Ethical, Sustainability and Safety aspects

This chapter is a reflection on the bigger implications of the Intelligent Substation Maintenance Chatbot: the difference it will make on a personal and organisational level, impose legal and ethical responsibility, its relevance to sustainability, and the safety / security like precautions that it must be responsibly implemented. The institutional reporting style used in documentation of project is directed and demonstrated.

8.1 Social Aspects

Positive impacts

- Improves field productivity and decision support through the provision of technicians with real time and standards reinforced procedures (removes wastage of time because of manuals).
- Retains the institutional knowledge (minimizes the loss of skill with the departure of the senior personnel) and is a source of on-job training of junior technicians.

Negative / mitigations

- Difficulties within the demand of routine advisory functions - relieve by means of upskilling programs and the staff redistribution to more valuable functions.
- The problems of the digital-divide in the remote areas: provide knowledge packages as well as off-line/packed and easy-to-use interface in order to reduce barriers to access.

Who is responsible?

- Project owners (utility / deployer) are left with the operational duty of ensuring that content and field is properly configured, counseled and trained.
- The developers and the maintenance would keep the technical correctness and version control of embedded manuals and update the documents on time.

- Supervisory engineers have to reserve the right to make safety-critical choices, the output of chatbots must be consultative in its character, must not be directive in any way.

8.2 Legal Aspects

Data protection & compliance

- Treat record of chats and other personalised facts of any technician as sensitive: data-minimisation, security storage, retention and access policy. Consume the guidelines of data protection requirements defined in the laws of the country and organisation; include the consent/ notice feature within the UI.

Liability & disclaimers

- Sources of taking up should be indicated as advisory, and outputs. Retrieve sources query (retrieved sources - response): Those audit logs must be maintained to facilitate tracking of decision in case of an incident. Deployers will in SLA and in-house SOPs mark out rules of liability.

Regulatory considerations

- Ensure that the standards of the industry that are used as references (e.g., transformer and maintenance standards discussed in the basis of knowledge) are adhered to. Periodically have a re-certification policy of the embedded content.

8.3 Ethical Aspects

Core principles

- Speaking of which, human control should deal with all actions on the fields prescribed by the chatbot.
- Transparency provides provenance (what manual/page the answer was got) and confidence scores/ uncertainty scores.
- Equity and non-discrimination: no instance of discriminating guidance- embedded documentation practices ought to be of accepted and usual practice.

Risks and mitigations

- **Over-reliance:** technicians can take the recommendations of the chatbots without verifying them. By means of training, UI prompts (check with senior) as well as high-risk procedure checks can be mitigated.
- **Misuse / dishonesty:** an effort to actively disable or alter logs (when intending to cover up their errors) ought to be hindered by employing access regulations, tamper-evident logs and audit paths; disciplinary and criminal rules need to exist.

Professional responsibility

- The operators and the engineers are supposed to put the safety of the population and their work places first rather than convenience. Any system and policy design should not accept unethical or unlawful request (e.g. bypassing safety checks).

8.4 Sustainability Aspects

Environmental benefits

- **Resource efficiency**: precise maintenance can minimize unwarranted part replacements and waste, increase asset life, as well as reduce material use.
- **Intermediate field travel:** remote guidance enhances with lower field travel means that the technician travel is minimized, with reduction in transport emissions.

Design considerations

- Considered thoughts with a preference towards lightweight compute (when available), efficient vectors stores and premeditated re-indexing to reduce repeated and usually heavy computations. Common path Memes are chosen to use to route around clouds and reduce the number of calls to the cloud and energy used.

Social dimension

- By preserving institutional knowledge -or indirect contribution toward sustainable development goals (industry, decent work, responsible production), the system assists in the promotion of sustainable institutions and local capacity building.

8.5 Safety & Security Aspects

Operational safety

- Chatbot cannot and does not in any way replace the necessary safety measures- any response that indicates that there is a field operation should refer to the operation of lockout/tagout procedures and permit procedures. The human-in-the-loop is mandatory in the safety-critical activities.

Cybersecurity

- Secure the endpoints and the data in transit/storage using TLS, powerful authentication, RBAC and secrets management. Patches and vulnerability checker financing. Track and monitor the unusual usage patterns.

Fail-safe & resilience

- **Design fallbacks**:when retrieval or LLM services are not online, the UI has to recover, using an offline quick-ref (describing SOPS), which is available. Use rate limiting, circuit breaking and back-off in API clients in order to prevent cascading failures.

Testing & validation

- Quite frequently compare answers to revised standards and perform red-team tests to seek unsafe outputs. Have a documented incidence response and a rollback plan.

Chapter 9
Conclusion

The Intelligent Substation Maintenance Chatbot successfully uses a Retrieval-Augmented Generation (RAG) architecture to provide correct standards based responses to maintenance queries of substations. The system achieves the major goals of having authoritative manuals, SOPs, and IEC/CBIP guidelines embedded in a semantic vector database and integrated with an LLM to achieve the desired results: reliable query classification, accurate context retrieval, safe response generation, and real-time technician support. The comparison of the evaluation results revealed high scores of retrieval accuracy, low latency, and high factual correctness which proves that the chatbot is successful in filling the gap between complex maintenance documentation and field-level decision making.

It is also shown in the project that AI-assisted maintenance could potentially contribute in a meaningful way to productivity of technicians, less ambiguity, and retention of expertise, which are the modern trends in substation conversational agents research. It will be enhanced with real-time SCADA data integration, multilingual support, more diagnostic reasoning based on knowledge graphs, and run on edge environments to enhance privacy and resiliency in the future. In sum, the system provides a firm basis of smart, reliable and scalable maintenance support in contemporary substations.

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Base Paper

Ni, M. et al., 2024. ChatGrid: Intelligent Knowledge Q&A for power dispatching control based on large language models and retrieval-augmented generation. *Energy Informatics*, 7(1), pp.1–15.

Appendix

i. Data Sheets / Technical Specifications

- **Gemini / LLM Model:** Cloud-hosted large language model with safety-tuned reasoning; supports grounded generation through RAG.
- **HuggingFace Embedding Model (BGE / Sentence Transformer):** 768-dimensional embeddings, optimized for semantic similarity and technical domain retrieval.
- **ChromaDB Vector Store:** Persistent local vector database; cosine similarity search; supports metadata filtering.
- **FastAPI Backend:** Python-based ASGI framework; average latency 1.2–1.8 seconds; JSON-validated endpoints.
- **Streamlit Frontend:** Supports real-time query display; lightweight rendering pipeline.
- **Python Environment:** Python 3.10; core packages include LangChain, Uvicorn, Pydantic, Chroma, HuggingFace, and Streamlit.

ii. Publications / Documentation

- **Conference Submission**
  Status: N/A.
- **Scopus/WoS URL:** N/A

iii. Project Report – Similarity Report

- **Similarity Index:** N/A
- **Tool Used:** N/A

iv. Datasets Used

- **Embedded Manuals Dataset:**
  - Transformer maintenance procedures (IEC 60076 excerpts).
  - Circuit breaker O&M guidelines.
  - CT/PT test procedures and safety rules.

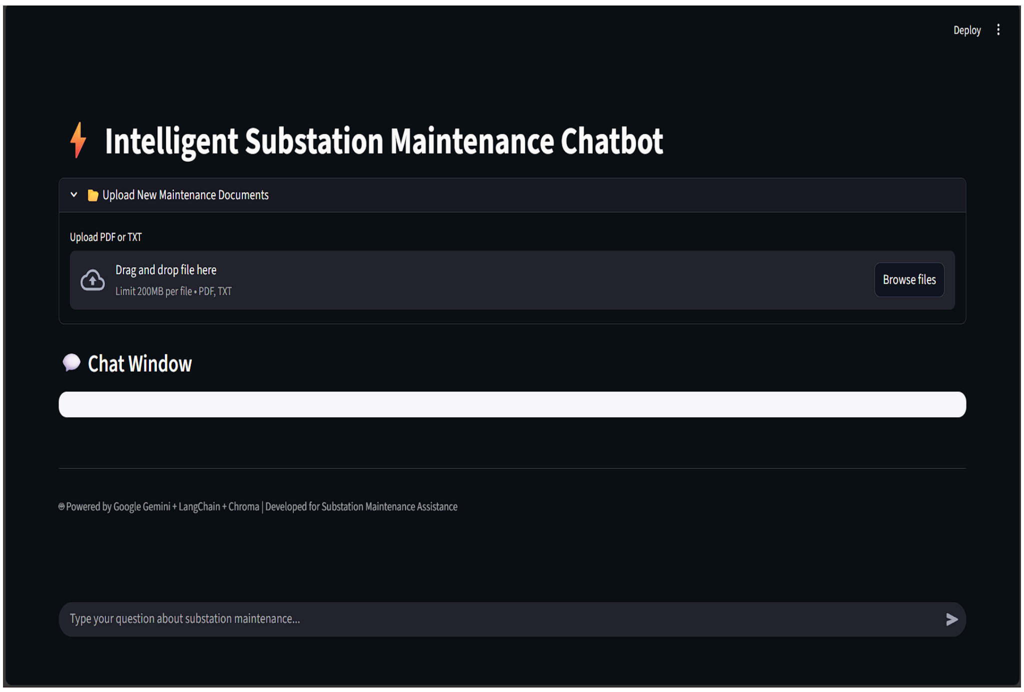
- Protection system troubleshooting notes.
- CBIP maintenance manuals.

- **Processed Dataset Format:**
  - Total Chunks: 1,000–1,800 text chunks
  - Average Chunk Size: 600–900 characters
  - Embedding Dimension: 768
- **Simulated Testing Dataset:**
  - 80 benchmark queries (transformer, breaker, CT/PT, relay, diagnostics).
  - 50 simulated fault scenarios for retrieval evaluation.

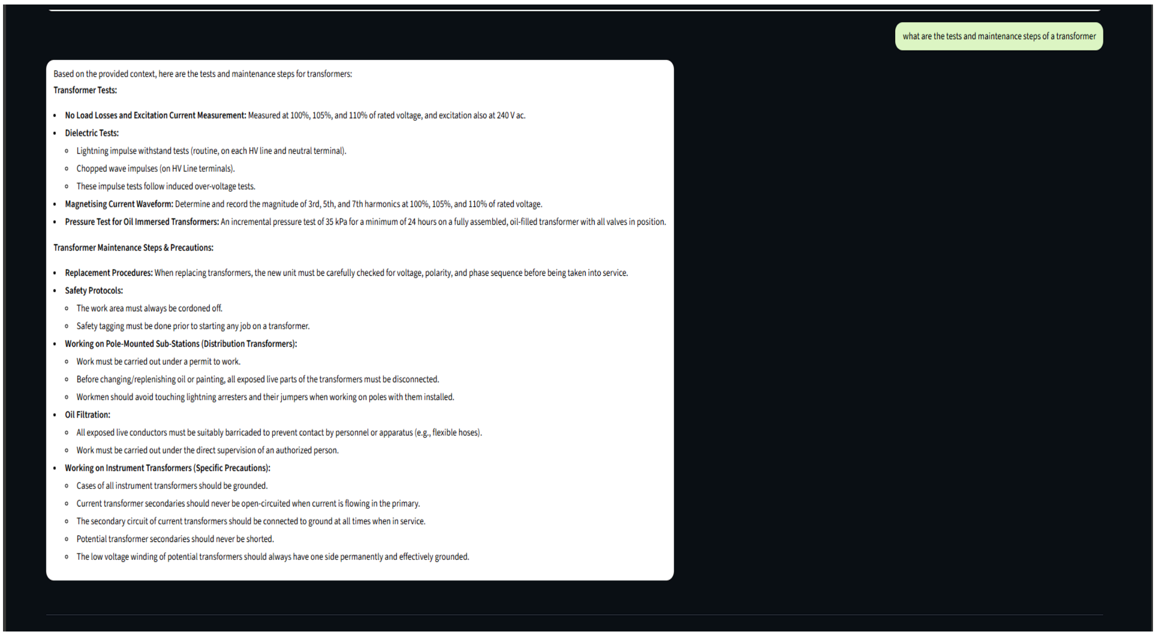
v. Auxiliary Documents

- Requirements Specification Document
- System Design Diagrams (RAG pipeline, backend microservices flow, UI flow)
- Testing Sheets (unit tests, integration tests, validation results)
- Software Bill of Materials (Python packages, versions)
- GitHub repository - https://github.com/substation-chatbot/substation-chatbot

vi. Few Images of Project



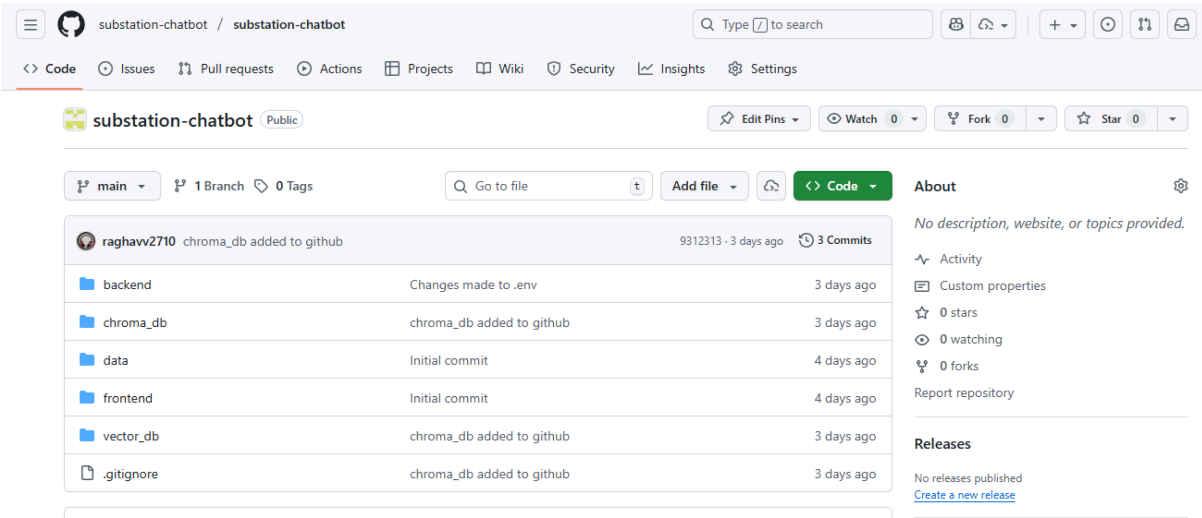
a. Chatbot Home Interface



b. Retrieval-Augmented Response Example

```python
# ==================================================
# RAG Answer Function
# ==================================================
def get_answer(query):
    """Retrieve context and ask Gemini to generate a response."""
    docs = retriever.invoke(query)
    context = "\n\n".join([doc.page_content for doc in docs])

    prompt = f"""
You are a Substation Maintenance Expert.
Use the following context to answer the question clearly and concisely.

Context:
{context}

Question: {query}
"""

    try:
        response = llm.invoke(prompt)
        return response.content if hasattr(response, "content") else str(response)
    except Exception as e:
        return f"⚠️ Error: {e}"
```

c. Query Classification Output Code Snippet



d. Github Repository Image